Overview

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2 Foundations

A **population** with N elements can be represented by the **finite** index set

$$U = \{1, 2, \ldots, N\}.$$

- The notation U is derived from "universe."
- Each label i ($i \in U$) can be assigned **uniquely** to one unit.

The elements $i \in U$ have attributes or characteristics that can be recorded by **variables**.

- Unless stated otherwise, we assume that all variables are *real-valued*.
- They are **not random variables**; their values are fixed.

Let y denote the variable of interest, with value y_i for population element $i \in U$. The most important (population) **characteristics** are

$$t_y = \sum_{i \in U} y_i \qquad \text{(population total)}$$

$$\bar{y}_U = \frac{1}{N} \sum_{i \in U} y_i = \frac{t_y}{N} \qquad \text{(population mean)}$$

$$\sigma_y^2 = \frac{1}{N-1} \sum_{i \in U} (y_i - \bar{y}_U)^2 \qquad \text{(population variance).}$$

- Short notation: t and σ^2

2.1 Sample and sampling design

Definition 1 (Sample).

A **sample** s is a **subset** of U, formally $s \subseteq U$.

- Sets are unordered objects (they have no additional structure or ordering).
- Conceptual model: drawing numbered balls from an urn **without replacement** (each element can occur at most once).
- *Alternative definition*
  - A sample (of size n) *with replacement* can be viewed as a **sequence** or **multiset** $s = \{i_1, i_2, \ldots, i_n\}$, where $i_k \in U$ for $k = 1, \ldots, n$.
  - In our finite-population setting, **selecting the same unit** more than once does not reveal an additional population value.

- We distinguish between **probability** and **nonprobability** sampling methods.
- In what follows, we focus on **probability sampling** and probability samples **without replacement**.

Set of all possible samples

The **set of all possible** samples from U is defined as

$$\mathscr{S} := \{s : s \subseteq U\}.$$

It contains:

- the empty set $\emptyset$ and the improper subset U; hence, $\mathscr{S}$ is the power set and contains 2^N elements,
- only samples in which each element can occur *at most once*.

Example 2 (U3).

For $U = \{1,2,3\}$, $\mathscr{S}$ contains the following 2^3 subsets:

$$s_0 = \emptyset,$$
$$s_1 = \{1\}, \qquad s_2 = \{2\}, \qquad s_3 = \{3\},$$
$$s_4 = \{1,2\}, \qquad s_5 = \{1,3\}, \qquad s_6 = \{2,3\}, \qquad s_7 = \{1,2,3\}.$$

The set of all possible samples from U with **fixed sample size** n is given by

$$\mathscr{S}_n = \{s : s \subset U \text{ and } |s| = n\}, \qquad (n > 0),$$

and contains

$$|\mathscr{S}_n| = \binom{N}{n} = \frac{N!}{n!\,(N-n)!}$$

distinct elements (i.e., individual samples).

For the discussion that follows, it is useful to introduce a **quantity associated with the elements** $i \in s$.

Definition 2 (Sample inclusion indicator).

Let s be given. The **sample inclusion indicator** is the indicator variable

$$\mathbb{1}\{i \in s\} := \begin{cases} 1 & \text{if } i \in s, \\ 0 & \text{otherwise,} \end{cases}$$

indicating whether an element $i \in U$ is included in the sample s (short notation: $\mathbb{1}_i$).

Remarks:

- Inclusion indicators allow us to write sample sums as sums over the entire population: $\sum_{i \in s} y_i = \sum_{i \in U} \mathbb{1}_i y_i$.
- This representation is particularly useful for derivations and proofs.

2.2 Probability sample

Example 3 (Simple random sampling without replacement, SRS).

Let U be a population of size N. We want to select a (proper) sample s of fixed size n **at random** from U (without replacement, $0 < n < N$).

- There are $|\mathscr{S}_n| = \binom{N}{n}$ possible samples.
- Each sample s has the same probability

$$\frac{1}{|\mathscr{S}_n|} = 1 \Big/ \binom{N}{n}$$

  of being selected.
- This defines a **discrete uniform distribution** (i.e., a probability distribution) on $\mathscr{S}_n$.

Definition 3 (Probability distribution).

Let $\mathscr{S}$ denote the (nonempty) set of all samples (sample space). We define a real-valued **function** $\mathbb{P} : \mathscr{S} \to [0,1]$ that assigns a **probability** to each **elementary event** $\{s\}$, where $\mathbb{P}$ satisfies the following properties:

1. $\mathbb{P}(\{s\}) \geq 0$, for every $s \in \mathscr{S}$, (nonnegativity)

2. $\sum_{s \in \mathscr{S}} \mathbb{P}(\{s\}) = 1$. (normalization)

3. $\mathbb{P}(E) := \sum_{s \in E : s \in \mathscr{S}} \mathbb{P}(\{s\})$, for every subset E of $\mathscr{S}$. (additivity)

We can now formulate three *fundamental* definitions of classical survey sampling theory: the sampling design, the inclusion probability, and the probability sample.

Definition 4 (Sampling design).

The mapping $p : \mathscr{S} \to [0,1]$, defined by $s \mapsto \mathbb{P}(\{s\})$, is called a **sampling design** (or simply design; also sampling plan).

Remarks:

- We use p for the design and write $p(s)$ for the probability of sample s under design p.
- To distinguish between designs, we use subscripts, e.g., p_{SRS}.

The **design**

- is the **central element** of classical survey sampling theory;
- determines the **sampling distribution** and therefore the stochastic properties of estimators (see Section 2.4).

Why is the design central to survey sampling theory?

To answer this question, consider the **freedom** we have in specifying a design. A design allows us to:

1. choose an **arbitrary** set of possible samples to which we assign positive probability;
2. distribute the total probability mass **arbitrarily** across the samples in this set, subject to $\sum_{s \in \mathscr{S}} p(s) = 1$.

We are **free**, within the framework of Definition 4, to specify a design for our purposes.

- The only **constraints** arise from considerations of **practicality** and **optimality** (e.g., estimators with smaller variance).
- For example, we could specify a design that assigns a different probability to every possible sample (see Chapter 5).

Example 4 (U3, cont. of Ex. 2).

Consider the design:

$$p_{U3}(s_k) = \begin{cases} 1/3 & \text{for } k = 4, 5 \text{ and } 6 \\ 0 & \text{for } k = 0, 1, 2, 3 \text{ and } 7. \end{cases}$$

Only samples of size $n = 2$ (i.e., $s_4 = \{1, 2\}$, $s_5 = \{1, 3\}$, and $s_6 = \{2, 3\}$) have a positive probability of being selected.

In addition to the probability $p(s)$ of a sample s, we are interested in the probability that an **element** $i \in U$ is included in s.

Definition 5 (Inclusion probability).

Let p be given. The probability that *element* i is included in sample s is called its (first-order sample) **inclusion probability** and is given by

$$\pi_i := \mathbb{P}(i \in s) = \sum_{s:i \in s} p(s) \qquad (i \in U),$$

where $\{s : i \in s\}$ is the set of all samples s that contain element i.

Example 6 (U3, cont. of Ex. 4).

Let $p_{U3}(s_k)$ be the design from Example 4. Under this design, the samples

$$s_4 = \{1,2\}, \qquad s_5 = \{1,3\} \qquad \text{and} \qquad s_6 = \{2,3\}$$

are the only possible samples with positive probability $p_{U3}(s_k) = 1/3$, $k = 4,5,6$. It follows that the inclusion probabilities are $\pi_i = 2/3$ for every $i \in U$.

Example 7 (SRS, cont. of Ex. 3).

Let U be a population of size N. Under **simple random sampling** (SRS) of size n $(0 < n < N)$ from U,

$$\pi_i = \frac{n}{N}, \qquad \text{for} \quad i \in U.$$

- The inclusion probabilities π_i $(i \in U)$ **alone** do not fully characterize a design.
  - $\Rightarrow$ Besides SRS, for example, there are other designs that also yield inclusion probabilities $\pi_i = n/N$.
- A **necessary and sufficient condition** for SRS is that every possible sample s has the same probability $p_{\mathrm{SRS}}(s)$ of being selected.
- Using the inclusion probabilities, we can now give a *common* definition of a probability sample (see, e.g., Wikipedia).

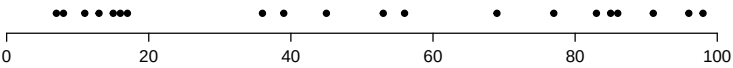
Definition 6 (Probability sample).

In a **probability sample**, every element $i \in U$ has a **known** inclusion probability $\pi_i > 0$.

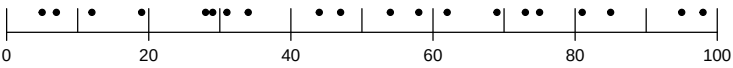
Remarks

- "Known" means that π_i is available for every $i \in U$ before the sample is selected.
- Using the π_i **together with knowledge of how** the sample is to be selected, or was selected, the design p can generally be (re-)constructed. This is essential for statistical inference.

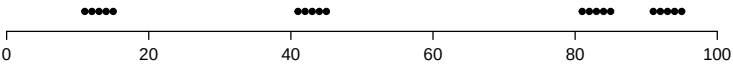
Simple random sampling without replacement (SRS)



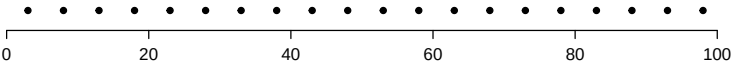
Stratified SRS (2 elements from each stratum)



Simple cluster sample (4 of 20 clusters)



Systematic sample



Summary

1. Population $U = \{1, \ldots, N\}$
2. Sample space $\mathscr{S}$ and sampling design p

$$\mathscr{S} \subseteq 2^U, \qquad p : \mathscr{S} \to [0,1], \qquad \sum_{s \in \mathscr{S}} p(s) = 1$$

with

$$\mathbb{P}_p(A) = \sum_{s \in A} p(s), \qquad A \subseteq \mathscr{S} \qquad \text{(probability measure induced by } p\text{)}$$

3. Random sample S and realization s

$$S \in \mathscr{S}, \qquad \mathbb{P}_p(S = s) = p(s), \qquad s \in \mathscr{S}$$

4. Population elements i and inclusion

$$i \in U, \qquad \mathbb{1}_i = \mathbb{1}\{i \in S\}, \qquad \pi_i = \mathbb{P}_p(i \in S)$$

2.3 Sample selection

- For large populations, it is often impractical to enumerate all possible samples.
- A **sampling scheme** (sampling algorithm) allows us to select a sample without enumerating all possible samples.
  - A **sequential** sampling scheme is a selection procedure applied to the list of units in U.
  - For a fixed-size design with sample size n, the sample can be selected using a **draw-by-draw sampling scheme**. At each of the $1, 2, \ldots, n$ steps of the procedure, one unit is selected.
- Many schemes and algorithms exist, e.g., eliminatory sampling and rejective algorithms; see Tillé (2006, Chapter 3).

2.4 Randomization inference

- **Randomization inference** concerns the **theoretical properties** of estimators under **repeated sampling** under the same conditions.

  $\Rightarrow$ The main object of interest is the **sampling distribution** of an estimator.

  $\Rightarrow$ In particular, it allows us to **quantify the precision of an estimator**.

- Let θ_U denote the (population) characteristic of interest (e.g., population mean or total), and let $\widehat{\theta}$ be an estimator of θ_U.

- We write $\widehat{\theta}(s)$ to emphasize that it is a **statistic**.

Definition 7 (Expected value and variance).

Let p be given. The **expected value** and **variance** of the statistic $\widehat{\theta}(s)$ are defined as

$$\mathbb{E}_p(\widehat{\theta}) := \sum_{s \in \mathscr{S}} p(s)\, \widehat{\theta}(s), \qquad \text{(expected value)}$$

$$\mathrm{var}_p(\widehat{\theta}) := \sum_{s \in \mathscr{S}} p(s)\, \{\widehat{\theta}(s) - \mathbb{E}_p(\widehat{\theta})\}^2. \qquad \text{(variance)}$$

Remarks:

- Expected value and variance are taken **over all possible samples** $s \in \mathscr{S}$ under the design p; we therefore write $\mathbb{E}_p$ and var_p.
- Recall that y_i ($i \in U$) are fixed quantities, not random variables. The sampling distribution arises **exclusively** from the random selection of the sample.

- Expected value and variance can be used to define **additional theoretical measures** of a sampling distribution.

$$\text{bias}_p(\widehat{\theta}) := \mathbb{E}_p(\widehat{\theta}) - \theta_U \qquad \text{(bias)}$$

$$\text{SE}_p(\widehat{\theta}) := \sqrt{\text{var}_p(\widehat{\theta})} \qquad \text{(standard error)}$$

$$\text{CV}_p(\widehat{\theta}) := \frac{\text{SE}_p(\widehat{\theta})}{\mathbb{E}_p(\widehat{\theta})}, \quad \text{if} \quad \mathbb{E}_p(\widehat{\theta}) > 0 \qquad \text{(coefficient of variation)}$$

$$\text{MSE}_p(\widehat{\theta}) := \text{var}_p(\widehat{\theta}) + \left\{\text{bias}_p(\widehat{\theta})\right\}^2 \qquad \text{(mean squared error)}$$

- An estimator $\widehat{\theta}$ is **unbiased** for θ_U if

$$\text{bias}_p(\widehat{\theta}) = 0 \qquad \text{equivalently} \qquad \mathbb{E}_p(\widehat{\theta}) = \theta_U.$$

- Let $\widehat{\theta}_1$ and $\widehat{\theta}_2$ be two unbiased estimators of θ_U under the same sampling design p. Estimator $\widehat{\theta}_1$ is **more efficient** than $\widehat{\theta}_2$ if

$$\text{var}_p(\widehat{\theta}_1) < \text{var}_p(\widehat{\theta}_2).$$

- In most cases, an **estimator** $\widehat{\text{var}}_p(\widehat{\theta})$ of the variance $\text{var}_p(\widehat{\theta})$ can be constructed using only the sample data.

To compare different sampling designs and their estimators, it is useful to introduce the concept of a strategy. Let θ_U denote the characteristic of interest.

Definition 8 (Strategy).

A **strategy** is a pair $(p, \widehat{\theta})$ consisting of a sampling design p and an estimator $\widehat{\theta}$ of θ_U.

Remarks:

- Example: $(p_{\text{SRS}}, \bar{y})$ denotes the sample mean $\bar{y} = (1/n) \sum_{i \in s} y_i$ as an estimator of the population mean $\bar{y}_U$ under SRS of size n.
- Let $(p_1, \widehat{\theta}_1)$ and $(p_2, \widehat{\theta}_2)$ be two different (otherwise unspecified) strategies. We could, for example, use mean squared error to compare them.

2.5 Asymptotics

- We have made **no distributional assumptions** about the variable of interest y or its population values y_i $(i \in U)$. Statistical inference therefore relies on **approximation methods**.
- In classical statistics, the **central limit theorem** (CLT) can be used to construct confidence intervals.
- The notion of asymptotics may seem incompatible with survey sampling from *finite* populations. Indeed, it is impossible to let n tend to infinity while keeping N fixed.
- We therefore need a **more elaborate asymptotic framework** in which $n_k, N_k \to \infty$ as $k \to \infty$.

Satz 1 (Hájek, 1960).

Suppose that $n_k/N_k \to f \in (0,1)$ *as* $k \to \infty$. *If the elements of the sequence* $(y_i)_{i\in\mathbb{N}}$ *satisfy*

$$\lim_{k\to\infty} \frac{\max_{i\in U_k}(y_i - \bar{y}_{U_k})^2}{\sum_{i\in U_k}(y_i - \bar{y}_{U_k})^2} = 0, \tag{N}$$

then, as $k \to \infty$,

$$\frac{\bar{y}_{s_k} - \bar{y}_{U_k}}{\widehat{\mathsf{SE}}_p(\bar{y}_{s_k})} \xrightarrow{d} N(0,1).$$

- $N(0,1)$ denotes the normal distribution with mean 0 and variance 1, and $\xrightarrow{d}$ denotes convergence in distribution (convergence in law).
- Condition (N) is called the Noether condition. It ensures that the variance is not dominated by a single observation y_i.

Confidence intervals

- The CLT allows us to construct a **confidence interval** (CI) for the population mean $\bar{y}_U$ based on its SRS estimator $\bar{y}$, provided that n is sufficiently large.

- A $100(1-\alpha)\%$ confidence interval can be constructed as

$$\bar{y} \pm z_{1-\alpha/2} \widehat{\mathsf{SE}}_p(\bar{y}),$$

where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of the $N(0,1)$ distribution.

Remarks

- Cassel et al. (1977, Chapters 1.2 and 1.3) define a sample in two different ways: the "set case" (corresponding to our Definition 1) and the "sequence case" (a sample defined as a sequence or multiset).
- Limit arguments and asymptotic properties of estimators and strategies are discussed in detail in Fuller (2009, Chapter 1.3).
- The theorem by Hájek cited above is a special case of Theorem 3.1 in Hájek (1960). In this special case, the Lindeberg condition is equivalent to the stated Noether condition; see Hájek (1960, Remark 3.3).

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Version of October 19, 2026